

ANTIMICROBIAL STUDIES ON FRACTIONS AND PURE COMPOUNDS OF *CALOPHYLLUM INOPHYLLUM* LINN.

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ABSTRACT

An evergreen medicinal plant, *Calophyllum inophyllum* Linn. (Guttiferae), is cultivated as an ornamental plant through out Pakistan and medicinally used as antiseptic, astringent, expectorant, analgesic, diuretic and purgative. The chemical investigations revealed organic compounds viz. cholesterol, friedelin, canophyllol, canophyllic acid, inophynone, isoinophynone, 1,4,5,6-tetrahydroxy-2-oxo-cycloheptene and biological studies (anti-bacterial and anti-fungal activities) of fractions (*n*-butanol, ethanol, chloroform and aqueous) and pure compounds (friedelin, canophyllol, canophyllic acid and inophynone) showed significant and non-significant results against *Staphylococcus aureus*, *Streptococcus pyogenes*, *Corynebacterium diphtheriae*, *Bacillus subtilis* (Gram +ve), *Escherichia coli*, *Salmonella typhi*, *Shigella boydii*, *Klebsiella pneumonia*, *Proteus mirabilis*, *Pseudomonas aeruginosa* (Gram -ve), *Epidermophyton floccosum*, *Trichophyton rubrum*, *T. schoenleinii*, *Pseudallescheria boydii*, *Candida albicans*, *Aspergillus niger* (human pathogens), *Microsporium canis*, *Trichophyton simii*, *T. metagrophyton* (animal pathogens), *Fusarium oxysporum* var. *lycopersici*, *F. solani* var. *lycopersici*, *Macrophomina phaseolina* (plant pathogens).

Key words:

Antimicrobial activity, *Calophyllum inophyllum* Linn., Guttiferae, Xanthonenes, Coumarins.

INTRODUCTION

Calophyllum inophyllum Linn. (Guttiferae) is an evergreen plant and widely cultivated through out Pakistan as an ornamental. All parts of it are used in medicine as antiseptic, astringent, expectorant, analgesic, diuretic and purgative (Nasir & Ali, 1973; Said, 1969; Wagner *et al.*, 1977; Wijesckera, 1991; Perry, 1980). The balsams from bark is used as a cicatrisant, infusion or decoction of leaves as an eye remedy in Asian medicine (Iinuma *et al.*, 1994). Guttiferaceous species are a rich source of xanthonenes, biflavonoids, benzophenones, neoflavonoids and coumarin derivatives (Iinuma *et al.*, 1994). A literature survey on chemical-biological activity relationship indicates the cytotoxic, anti-tumor, anti-inflammatory, anti-fungal, enhancement of choline acetyltransferase, anti-HIV activity and inhibition of lipid peroxidase of

xanthenes and coumarins (Hosttetmann & Hosttetmann, 1989; Sordat-Diserens *et al.*, 1992; Fukuyama *et al.*, 1991; Minami *et al.*, 1993; Boyd *et al.*, 1993). The present studies was carried out with the object to search biologically active principles related to xanthone and coumarins, and this research article deals with anti-bacterial (gram positive and negative strains) and anti-fungal activities (human, animal and plant organisms) of crude extract's fractions and pure compounds isolated from it viz. friedelin, canophyllol, canophyllic acid, inophynone. No such activity has been reported prior to our communication.

EXPERIMENTAL

Materials and Methods:

The anti-microbial activity tests were initially carried out on Gram positive & negative bacteria and human, animal and plant pathogenic fungi. The bacterial strains used in this work were *Staphylococcus aureus*, *Streptococcus pyogenes*, *Corynebacterium diphtheriae*, *Bacillus subtilis* (Gram +ve), *Escherichia coli*, *Salmonella typhi*, *Shigella boydii*, *Klebsiella pneumonia*, *Proteus mirabilis*, *Pseudomonas aeruginosa* (Gram -ve) and the fungal species *Epidermophyton floccosum*, *Trichophyton rubrum*, *T. schoenleinii*, *Pseudallescheria boydii*, *Candida albicans*, *Aspergillus niger* (human pathogens), *Microsporum canis*, *Trichophyton simii*, *T. metagrophyton* (animal pathogens), *Fusarium oxysporum* var. *lycopersici*, *F. solani* var. *lycopersici*, *Macrophomina phaseolina* (plant pathogens).

The anti-bacterial activity was determined by well diffusion technique (Carran *et al.*, 1987). The test samples and reference (standard) drugs i.e. amoxicillin (H₂H)₃, ampicillin (H₂O)₃ were taken in different concentration in DMSO. The experimental plates were incubated at 37°C for 24 hours and the zone of inhibitions were measured in mm and compared with positive controls. The results are given in Table 1a, 1b, 2a and 2b.

The anti-fungal activity was determined on Sabouraud Dextrose Agar (SDA) plates. Four different fractions i.e. *n*-butanol, ethanol, chloroform, aqueous fractions and pure compounds, friedelin, canophyllol, inophynone were examined on 12 different pathogenic fungal species of human, animal and plant sources.

The classical method was used for the preparation of screw up cap tubes slant. SDA was prepared by mixing 32.5 gram Sabouraud, 4% glucose agar and 4 gram agar-agar in 500 ml distilled water. The screw caps tubes with media were autoclaved at 121°C for 15 minutes. The tubes were allowed to cool (50°C) and non-solidified Sabouraud dextrose agar media were poisoned with 66.6 ml of samples pipetted from stock solution. The media of final concentration i.e. 400 mg/ml were used for the

determination of anti-fungal activity of fractions (*n*-butanol, ethanol, chloroform, aqueous) and pure compounds (friedelin, canophyllol, inophynone). For optimum growth of organisms the usual incubation period 27-29°C temperature and 7-10 days were used. For comprehensive results see Table 3a, 3b, 3c.

RESULTS AND DISCUSSION

Ethanol, *n*-butanol, chloroform and aqueous fractions of the leaves of *C. inophyllum* Linn. were used for antibacterial activity tests in concentration (100 µg/100 ml). The antibacterial activities of these fractions were also compared with the standard drugs such as ampicillin and amoxicillin.

The ethanol fraction showed moderate activity against two Gram +ve organisms i.e. *Staphylococcus aureus* and *Corynebacterium diphtheriae* while the activity against *Streptococcus pyrogene* and *Bacillus subtilis* was found insignificant. This extract did not exhibit any activity against test organisms of Gram -ve characters.

The *n*-butanol fraction was found insignificant against all Gram +ve and, -ve strains except *Escherichia coli* which showed significant activity. Significant activity was not found in aqueous fraction against all Gram +ve and -ve organisms used in testing, whereas chloroform fraction showed significant activities specially against various strains of Gram +ve bacteria such as *C. diphtheriae* (excellent), *S. aureus* (excellent), *S. pyrogene* (excellent) in 100 µl concentration when compared to control. However, this fraction was found totally inactive against all Gram -ve organisms which were used for its antibacterial activity (Table 1a, 1b).

Pure isolated compounds canophyllic acid, canophyllol, friedelin and inophynone (a new compound) were also subjected to antibacterial activity in concentration (100 µg/100 µl) against certain Gram +ve (*Staphylococcus aureus*, *Streptococcus pyrogenes*, *Corynebacterium diphtheriae*, and *Bacillus subtilis*) and Gram -ve (*Escherichia coli*, *Salmonella typhi*, *shigella boydii*, *Klebsiella pneumonia*, *Proteus mirabilis*) strains.

Canophyllic acid exhibited moderate activity against Gram negative organism *Proteus mirabilis* only when compared with standard drugs (ampicillin and amoxicillin) in 100 µg/100 µl concentration while against other Gram negative and all Gram positive organisms no activity was observed (Table 2a, 2b).

Canophyllol was found rather better antibacterial agent than other isolated compounds against Gram positive strains, *Staphylococcus aureus* (moderate), *Corynebacterium diphtheriae* and *Bacillus subtilis*. The zone inhibiting activity against Gram negative strain *Klebsiella pneumonia* was observed as good, and excellent

against *Proteus mirabilis* when was compared with control in 100 µg/100 µl concentration, whereas activity against other organisms was found insignificant (Table 2a, 2b).

Friedelin showed good activity in 100 µg/ml µl concentration against two Gram positive strains, *Staphylococcus aureus* (moderate) and *Corynebacterium diphtheriae* (good) and there Gram negative strains such as *Salmonella typhi* (good) *Klebsiella pneumonia* (good) and *Proteus mirabilis* (moderate) on comparison with control (Table 2a, 2b).

Inophynone displayed good activity against *Staphylococcus aureus* when compared its activity with standards, ampicillin and amoxicillin in 100 mg/100 gl concentration. While rest of the Gram positive strains and all Gram negative strains which were used in this exercise were found inactive against inophynone (Table 2a, 2b).

The ethanol, *n*-butanol, chloroform and aqueous fractions of *C. inophyllum* Linn. and the isolated compounds from this source, friedelin, canophyllol and a new pyran type of compound, inophynone, were screened for anti-fungal activity against 12 fungal culture in 400 mg/ml media concentration.

The anti-fungal activity of fractions (ethanol, *n*-butanol, chloroform and aqueous) and the pure compounds; friedelin, canophyllol and inophynone, were determined by growing fungal cultures in Sabouraud Dextrose Agar (SDA) supplemented with test substance as well as without test substance. The ethanol fraction exhibited activity against *Trichophyton metagrophyte*, *Aspergillus niger*, *Candida albicans*, *Trichophyton semii* and *Microsporum canis* and negligible activity against other fungi.

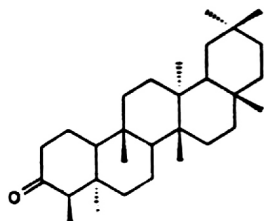
The *n*-butanol fraction was examined against different fungal cultures but the significant growth inhibition zones were observed in two species of *Trichophyton* i.e. *Trichophyton semii*, *Trichophyton metagrophyte* while no activity was found in other test organisms. The chloroform extract displayed an excellent activity against certain human pathogens such as *Pseudallescheria boydii*, *Aspergillus niger* and moderate activity against *Candida albicans* and *Trichophyton schoenleinii*. This fraction also exhibited a significant activity against various animal pathogens such as *Microsporum canis*. Chloroform fraction also showed moderate activity against *Macrophomina phaseolina* (plant pathogen) while against other fungal strains it was found insignificant (Table 3a).

In aqueous fraction a minimum significant activity was observed against some fungal strains such as *Trichophyton semii*, and *Microsporum canis*, and non significant against other fungal strains (Table 3a, 3b, 3c).

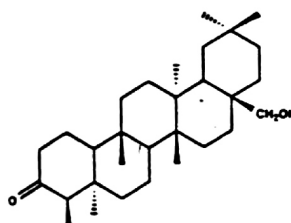
Friedelin belongs to the friedelin group of terpenes, displayed excellent activities against most of the human pathogen fungal strains such as *Aspergillus niger* and *Pseudallescheria boydii* while moderate activity was found against *Trichophyton schoenleinii* and *Candida albicans* which are common human pathogens. Certain animal pathogens were also tested against friedelin and found a greater degree of growth inhibition zones in the culture of *Microsporum canis* and *Macrophomina phaseolina* (plant pathogen) (Table 3a, 3b, 3c).

Canophyllol and inophynone did not exhibit any significant anti-fungal activity against any fungal strain used in this exercise, although, a low activity was observed against *Trichophyton schoenleinii* (human pathogen) and *Macrophomina phaseolina* (Plant pathogen) only (Table 3a, 3b, 3c).

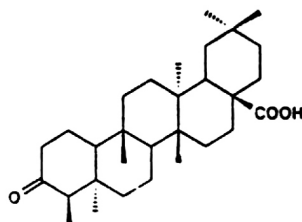
In view of above mentioned results of antibacterial/anti-fungal activity of friedelin, canophyllol, canophyllic acid, inophynone it is clear that the medicinal plant *Calophyllum inophyllum* possesses powerful bactericide/fungicide agents which were found effective against various human and animal pathogen. They could be employed clinically on-certain subcutaneous and opportunistic mycosis including infection of skin, lungs, eye and CNS and hypersensitivity etc, while in cutaneous mycosis like ring worm infection of hair and skin in animals, this plant also showed good results (Chopra et al., 1986; Kirtikar & Basu, 1981; Nadkarni, 1976). The antiseptic, cicatrisant, eye remedy activity of this plant maybe attributed to the above mentioned natural products. The research on this aspect is in progress.



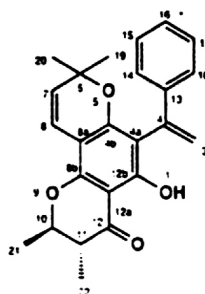
Friedelin



Canophyllol



Canophyllic acid



Inophynone

Table-1a
ANTIBACTERIAL ACTIVITY OF CRUDE EXTRACTS OF CALOPHYLLUM INOPHYLLUM LINN.
 (Gram Positive)

S. No.	Name of Organisms	Zone of Inhibition (mm) of Crude Extract 100 µg/100 µl				Standard Drugs	
		EtOH Extract	BuOH Extract	CHCl ₃ Extract	Aqueous Extract	Ampicillin 100 µg/100 µl	Amoxicillin 10 µg/100 µl
Gram Positive							
1	<i>Staphylococcus aureus</i>	6.5	-	6.0	-	24	25
2	<i>Streptococcus pyogenes</i>	14	-	6.5	-	23	20
3	<i>Corynebacterium diphtheriae</i>	6	-	7.0	-	23	25
4	<i>Bacillus subtilis</i>	13	-		-	23	25

- Test was not performed

Table-1b
ANTIBACTERIAL ACTIVITY OF CRUDE EXTRACTS OF CALOPHYLLUM INOPHYLLUM LINN.
 (Gram Negative)

S. No.	Name of Organisms	Zone of Inhibition (mm) of Crude Extract 100 µg/100 µl				Standard Drugs	
		EtOH Extract	BuOH Extract	CHCl ₃ Extract	Aqueous Extract	Ampicillin 100 µg/100 µl	Amoxicillin 10 µg/100 µl
Gram Negative							
1	<i>Escherichia coli</i>	.	7.5	.	.	.	.
2	<i>Salmonella typhi</i>	.	.	.	.	20	22
3	<i>Shigella boydii</i>	.	.	.	.	.	.
4	<i>Klebsiella pneumonia</i>	.	.	.	.	8	15
5	<i>Proteus mirabilis</i>	.	.	.	.	.	.
6	<i>Pseudomonas aeruginosa</i>	.	.	.	.	.	.

. Test was not performed

Table-2a
ANTIBACTERIAL ACTIVITY OF PURE COMPOUNDS FROM *C. INOPHYLLUM*
(Gram Positive)

Zone of Inhibition mm 100 µg/100 µl																
S. No.	Name of Organisms	Standard Drugs			Name of Control		Standard Drugs			Name of Control		Standard Drugs				
		Amoxicillin 100 µg/100 µl	Amoxicillin in 100 µg/100 µl	Cefuroxime 100 µg/100 µl	Clindamycin 100 µg/100 µl	Control me	Amoxicillin in 100 µg/100 µl	Amoxicillin in 100 µg/100 µl	Cefuroxime 100 µg/100 µl	Clindamycin 100 µg/100 µl	Control me	Amoxicillin in 100 µg/100 µl	Amoxicillin in 100 µg/100 µl	Cefuroxime 100 µg/100 µl		
Gram Positive																
1	<i>Staphylococcus aureus</i>	8.6	17.5	13.5	14.5	5.5	17.5	13.5	14.5	-	10	20	4.5	10	20	-
2	<i>Streptococcus pyogenes</i>	-	14.5	9.5	12.5	-	14.5	9.5	12.5	-	5.01	10	5.0	10	-	-
3	<i>Streptococcus agalactiae</i>	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	<i>Corynebacterium jeikeium</i>	3.8	13.0	11	18.5	4.5	13	11.0	18.5	-	8.5	11	12.5	8.5	11.0	12.5
5	<i>Bacillus subtilis</i>	-	11.5	10.0	13.0	4	11.5	10.0	13.0	-	-	-	-	-	-	-

- Test was not performed

Table-2b
ANTIBACTERIAL ACTIVITY OF PURE COMPOUNDS FROM *C. INOPHYLLUM*
(Gram Negative)

Zone of Inhibition mm 100 µg/100 µl																			
S. No.	Name of Organisms	Name of Compound			Standard Drugs			Name of Compound			Standard Drugs			Name of Compound			Standard Drugs		
		Phenoxymethyl penicillin 100 µg/100 µl	Ampicillin 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	Amoxycillin 100 µg/100 µl	Clavulanic acid 100 µg/100 µl	
Gram Negative																			
1	<i>Escherichia coli</i>	-	12.5	17.5	19.5	-	12.5	17.5	19.5	-	-	-	-	-	-	-	-	-	
2	<i>Salmonella typhi</i>	3.5	11.5	12.0	15.0	-	11.5	12.0	15.0	-	-	-	-	-	-	-	-	-	
3	<i>Shigella boydii</i>	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
4	<i>Moraxella pneumoniae</i>	4.0	11.0	10.5	15.0	3	11.0	10.5	15	-	-	-	-	-	-	-	-	-	
5	<i>Proteus mirabilis</i>	3	16.0	17.0	17.0	3.5	16.0	17.0	17.0	4.5	10	11	-	-	-	-	-	-	
6	<i>Pseudomonas aeruginosa</i>	-	11.5	18.0	12.5	-	11.5	18.0	12.5	-	-	-	-	-	-	-	-	-	

- Test was not performed

Table-3a
ANTIFUNGAL ACTIVITY OF CRUDE EXTRACTS AND PURE COMPOUNDS OF CALOPHYLLUM INOPHYLLUM LINN

S. No	Name of Organisms	Activity (Linear Growth mm)				Crude Extracts				Pure Compounds										Standards / Inhibition % Drugs																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
		Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial	Anti-fungal	Anti-bacterial																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1	<i>Aspergillus fumigatus</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+</

• Test was not performed

Table-3b
ANTIFUNGAL ACTIVITY OF CRUDE EXTRACT AND PURE COMPOUND OF CALOPHYLLUM INOPHYLLUM LINN
(Animal pathogens)

S. No	Name of Organisms	Activity (Linear Growth mm)						Crude Extracts						Pure Compounds						Standard Drugs	Inhibition %		
		BCH Extract at	Conc. at	Inhib. mm %	BCH Lysate at	Conc. at	Inhib. mm %	CHCl ₃ Extract at	Conc. at	Inhib. mm %	Aqueous Extract at	Conc. at	Inhib. mm %	Conc. in DMF %	Inhib. mm %	Conc. at	Control	Inhib. mm %	Inhib. at 100 µg/ml			Conc. at	Inhib. mm %
Animal Pathogens																							
1	<i>Microsporum canis</i>	40	90	56	65	70	7.1	20	85	75	37	90	50.3	25	87	72	63	67	5.77	92	81	3.58	100
2	<i>Trichosporon mentagrophytes</i>	40	92	57	15	65	77	90	82	3	60	92	60	-	-	-	-	-	-	-	-	-	-
3	<i>Trichosporon ashi</i>	41	91	55	16	60	84	10	100	80	66	91	29.55	68	102	92	80	76	18.42	40	38	5.25	100

- Test was not performed

Table-3c
ANTIFUNGAL ACTIVITY OF CRUDE EXTRACTS AND PURE COMPOUNDS OF CALOPHYLLUM INOPHYLLUM
LINN
(Plant pathogens)

S. No	Name of Organisms	Activity (Linear Growth mm)					Crude Extracts				Pure Compounds										Standard Drugs	Inhibition %
		BCH Extract	Conc at	Inhib mm %	BuOH Conc at	Center of	CHCl ₃ Conc at	Conc at	Inhib on %	Aqueous conc at	Conc at	Inhib mm %	Conc at	Inhib mm %	Conc at	Inhib mm %	Conc at	Inhib mm %				
(Plant pathogens)																						
1	<i>Fusarium oxysporum</i> var. <i>lycopersici</i> (tomato)	-	-	-	-	-	120	97	-23.7	-	-	-	111	87	-14.43	103	1.80	97	110	11.81	Bordeaux Mixture	100
2	<i>Fusarium solani</i> var. <i>lycopersici</i> (tomato)	-	-	-	-	-	115	121	5%	-	-	-	113	119	5.04	101	-12.12	90	94	-1.14	Bordeaux Mixture	100
3	<i>Macrophoma phaseolina</i>	-	-	-	-	-	30	80	62.5%	-	-	-	36	53	58.62	72	26.53	92	96	4.18	Bordeaux Mixture	100
4	<i>Rhizoctonia solani</i>	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Bordeaux Mixture	-

- Test was not performed

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